

3.4.2 Energy and rate of reaction

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 3			
Some reactions transfer energy to the surroundings so the temperature increases. Such reactions include combustion, oxidation and neutralisation.	Students are not required to know the term 'exothermic'.	5.5.1.1	4.7.3.3
Other reactions take in energy from the surroundings, so the temperature decreases. These reactions include dissolving ammonium chloride in water and reacting citric acid with sodium hydrogencarbonate.	Students are not required to know the term 'endothermic'. Suggested activity for TDA Compare the temperature changes caused by some reactions.	5.5.1.1	4.7.3.3
Outcome 4			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
The rate of a chemical reaction may be increased by increasing the temperature, increasing the concentration of reactants, increasing the surface area of solid reactants or by adding a suitable catalyst.	Students do not need to calculate rates of reactions, but should be able to compare relative rates by measuring the time for a reactant to be used up, the volume of a gas produced in a given time, the time for a solution to become opaque or coloured. Suggested activity for TDA Investigate how to make a chemical reaction go faster.	5.6.1.2	4.7.4.6